



[1]
✓ 0s

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
```



[8]
✓ 0s

```
data = pd.read_csv("decisiontree2.csv")
x= data[["tempMode","AQ","USS","CS","VOC","RP","IP","Temperature"]]
y= data["fail"]
x
```



...

	tempMode	AQ	USS	CS	VOC	RP	IP	Temperature
0	7	7	1	6	6	36	3	1
1	1	3	3	5	1	20	4	1
2	7	2	2	6	1	24	6	1
3	4	3	4	5	1	28	6	1
4	7	5	6	4	0	68	6	1
...	...	...	...	...	...	...	...	...
939	7	7	1	6	4	73	6	24
940	7	5	2	6	6	50	6	24
941	3	6	2	7	5	43	6	24
942	6	6	2	5	6	46	7	24



28 GB available

Files

- ..
- sample_data
- decisiontree2.csv

```
[1] ✓ 0s
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
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```
[2] ✓ 0s
data = pd.read_csv("decisiontree2.csv")
x= data[["tempMode","AQ","USS","CS","VOC","RP","IP","Temperature"]]
y= data["fail"]
data
```

...	footfall	tempMode	AQ	USS	CS	VOC	RP	IP	Temperature	fail
0	0	7	7	1	6	6	36	3	1	1
1	190	1	3	3	5	1	20	4	1	0
2	31	7	2	2	6	1	24	6	1	0
3	83	4	3	4	5	1	28	6	1	0
4	640	7	5	6	4	0	68	6	1	0
...	...	...	...	...	...	...	...	...	...	...
939	0	7	7	1	6	4	73	6	24	1
940	0	7	5	2	6	6	50	6	24	1
941	0	3	6	2	7	5	43	6	24	1
942	0	6	6	2	5	6	46	7	24	1



Files



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```
[1]
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
```

```
[9]
data = pd.read_csv("decisiontree2.csv")
x= data[["tempMode","AQ","USS","CS","VOC","RP","IP","Temperature"]]
y= data["fail"]
y
```

fail	
0	1
1	0
2	0
3	0
4	0
...	...
939	1
940	1
941	1
942	1

CO

decisiontree2.ipynb

☆

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RAM Disk

Files

📁 ..

📁 sample_data

📄 decisiontree2.csv

[19] ✓ Os

model= DecisionTreeClassifier(criterion='entropy')
model.fit(x,y)

DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy')

[20] ✓ Os

▶ sample=[[4,5,3,6,1,45,5,1]]

+ Code + Text

[21] ✓ Os

prediction=model.predict(sample)
print("Prediction: ",prediction)

Prediction: [1]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassi
warnings.warn(

[22] ✓ Os

if prediction == 0:
 print("SYSTEM WILL FAIL")
else:
 print("SYSTEM WILL FAIL")

SYSTEM WILL FAIL

Disk 87.25 GB available

